Sukhyun Han

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Research Interests

Domain Specific Accelerator (NPU) Architecture
- Deep Learning Accelerator Architecture
- Hardware-Software Co-Design Methodology

AI Acceleration
- Optimization for DL Accelerators
- LLM model Performance Engineering

Education

Ph.D. Course
Sungkyunkwan university, Suwon, South Korea (Aug. 2025 ~ Present)
Department of Electrical and Computer Engineering

Master's Degree
Sungkyunkwan university, Suwon, South Korea (Mar. 2024~ Aug. 2025)
Department of Semiconductor Convergence Engineering

Bachelor's Degree
Sungkyunkwan university, Suwon, South Korea (Mar. 2019 ~ Mar. 2024)
Department of Electronic and Electrical Engineering

Skills and Techniques

Programming Languages
- Digital system design with Verilog HDL
- Software programming with C, C++, Python, CUDA on Linux system

EDA Tool
- Synopsys Design Compiler
- Vivado FPGA Suite
- Cadence Virtuoso Suite

Honors and Awards

2025 Outstanding Research Award from SKKU Graduate School (Master's Program)
- Awarded the Excellence Award at DATE 2025 for my paper, "Zebra," which proposes a novel accelerator for Vision Transformers.

2022 AI Semiconductor design Competition (4th place prize)
- Design General HW Accelerator based on Row-stationary Systolic Array.
- Our team aimed to design and verify accelerator architectures that optimize dataflow and maximize throughput by row-stationary SA.

2022 Deep learning Hardware Design Competition (Popularity Award)
- Design tiny-YOLOv3 HW Accelerator with Verilog HDL.
- Our team aimed to design and verify accelerator architectures that maximize throughput by minimizing

off-chip memory access and leveraging maximum board resources. My work included implementation of Fused-Conv layer which is integrated with Pooling layer in CNN, reducing the access of On-chip SPM.

Major Publications

Avalanche: Optimizing Cache Utilization via Matrix Reordering for Sparse Matrix Multiplication Accelerator

- 2025 International Symposium on Computer Architecture (ISCA 2025)
- Gwangeun Byeon, Seongwook Kim, Hyungjin Kim, **Sukhyun Han**, Jinkwon Kim, Prashant Nair, Taewook Kang, Seokin Hong
- This paper addresses cache contention in sparse matrix multiplication accelerators with tiled outer product dataflow. The proposed architecture maximizes on-chip cache utilization by evicting early non-important data to external memory. My major contribution to this paperwork is to give advice about some details of the paper writing.

Zebra: Leveraging Diagonal Attention Pattern for Vision Transformer Accelerator

- 2025 Design, Automation & Test in Europe Conference & Exhibition (DATE 2025)
- **Sukhyun Han**, Seongwook Kim, Gwangeun Byeon, Jihun Yoon, Seokin Hong
- This paper proposes a novel accelerator for ViT. This paper reveals that the self-attention map of ViT models usually show the diagonal pattern, and it is possible to apply pruning with respect to the pattern. To leverage these sparse diagonal patterns, this paper proposes novel hardware architecture that only compute necessary vector operations in attention mechanism. I am the first author of this paper.

Leveraging Mobile Processors for ISAR Image Generation and Classification in Radar Platforms

- IEEE Access (Early Access)
- Seongwook Kim, **Sukhyun Han**, Woojin Cho, Youngjae Choi, Seungeui Lee, Youngseok Bae, Seokin Hong
- This paper explores using a low-power mobile application processor (AP) to automate and accelerate radar-based small target identification. We shows that a mobile AP can efficiently process raw ISAR data to generate and classify images in real-time, providing a lightweight solution for portable radar platforms. I designed and developed the device application for the demo platform. My role also included analyzing the ISAR image generation algorithm and conducting experiments to profile FFT and other operations.

Projects and Research Experiences

Research Fellow Aug. 2023 - Present

Computer Architecture and Systems Lab – COMPASS LAB SKKU

- Profiling and Performance Analysis for Transformer models.
- Teaching Assistant for Semiconductor System Engineering courses.
- Maintain and Manage GPU Computing server.

Comprehensive HW/SW Solution for Edge Self Supervised Learning Sep. 2023 - Present

COMPASS Lab

- Co-working with Hanwha Systems Inc.
- Evaluate the performance of Mobile AP embedded with NPU for ISAR-based object detection and classification task. My work included implementation of efficient ISAR-based algorithms on Mobile AP and design the control application.

Outlier Matters: A Statistical Analysis of LLM Tensor Distributions and Quantization Effects

International Conference paper

- 40th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC 2025)
- Taein Kim, Seongwook Kim, **Sukhyun Han**, Woojin Cho, Youngjae Choi, Youngseok Bae, Seokin Hong
- The paper proposes an "Outlier-Rescaled" quantization method, which uses a shift-based mechanism to

preserve the important outlier representations within a 4-bit format. This method significantly restores performance lost under INT4 quantization in LLMs without requiring additional hardware. I contributed to the design of a novel PE architecture for a shift-based mechanism.

Evaluating Performance of Modern GPU with Partitioned Last-Level Caches
International Conference paper

- 40th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC 2025)
- Jihun Yoon, Joonseong Hwang, **Sukhyun Han**, Sungbin Jang, Yoonho Jang, Seokin Hong
- This paper proposes an extended simulator that precisely models the A100 GPU architecture including network contention. This research reveals that partitioned cache designs can reduce L2 access latency by 1.17x, emphasizing the need for precise delay modeling in future GPU architecture evaluations. My major contribution to this paperwork is to give advice about some details of the paper writing.

NPU Technology for Accelerating Ladar-based Object Detection Sep. 2023 - Feb. 2025
COMPASS Lab

- Design NPU Microarchitecture including ISA and the programming model integrated RISC-V Instruction Set Simulator.
- My work included managing setup and documentation of Spike Simulator SDK and contributing to the development of RISC-V based extended ISA for edge NPU specialized for Self-Supervised Learning.

Samsung Computer Engineering Challenge 2023 Aug. 2023 - Nov. 2023
Personal Project

- Optimize LLaMav2 model Inference without accuracy drop.
- My work included implementing multi-GPU parallelization and advanced dataset sorting with dynamic padding reduction to minimize redundant computations.

Robust BGR and LDO Design against PVT Variations Jun. 2022 - Aug. 2022
CIRCUITS Lab in SKKU (Undergraduate Research Program)

- Design and evaluate AMP, BGR, and LDO circuits meeting specifications using 180nm TSMC process.
- My work included Improving and analyzing performance for PVT variation robustness by utilizing Thick PMos and Cascode PMos devices.

Implementation of Neural Network Computation using Low-Power CIM Circuit Mar. 2023 - Jul. 2023
Undergraduate Thesis

- Undergraduate Poster session at the 2023 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC).
- My work included Implementation an energy-efficient Computing-in-Memory circuit based on 8T SRAM by incorporating a Layer Recycling technique.

Real Time Face Recognition using MTCNN and FaceNet with TensorRT Sep. 2021 - Dec. 2021
Undergraduate Course Project

- Optimize MTCNN and FaceNet algorithms on Jetson Nano using CUDA for GPU/CPU co-processing.
- My work included real-time multi-face recognition with K-NN algorithm and enabled new face registration functionality, enhancing accuracy.